Zheng Ma

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Research Interests

Active Galactic Nuclei (AGN), High-Redshift Galaxy Evolution, Black Hole–Galaxy Coevolution, Large-Scale Structure

Education

Ph.D. in Physics, University of Arizona 2022–Present
Advisor: Eiichi Egami
Expected Graduation: 2027

M.Sc. in Physics, University of Copenhagen 2018–2021
Advisor: Subodh Patil
Thesis: *Gravitational Waves and Quasinormal Modes*

B.Sc. in Physics, Nanjing University 2014–2018
Advisor: Zuowei Liu
Thesis: *The Lithium-7 Problem in Big Bang Nucleosynthesis*

Publications

First-Author Papers

1. **Ma, Z.** et al. (2026), *A Quasar Hatching from a Buried Red Phase at* $z = 3.7$. arXiv:2607.22966, under revision.
2. **Ma, Z.** et al. (2026), *Undermassive Hosts of* $z = 4$–6 *AGN from JWST/NIRCam Image Decomposition with CONGRESS, FRESCO, and JADES.* arXiv:2601.15962, under revision.

Manuscripts in Preparation

1. **Ma, Z.** et al., *Infrared Variability of Little Red Dots.* In preparation.

Co-Authored Publications (Selected)

1. Zhu, Y. et al., including **Ma, Z.** (2026), *Early Metal-Enriched Baryon Cycling Before the Midpoint of Cosmic Reionization.* Nature Astronomy. DOI.
2. Zhu, Y. et al., including **Ma, Z.** (2026), *Low Lyα Visibility in Galaxy Overdensities: Reionization Topology and Neutral-Fraction Ceilings from DIVER over* $4.8 < z < 11$. arXiv:2608.19311.

3. Zhu, Y. et al., including **Ma, Z.** (2026), *PhotoIFU: NIRCam as a Photometric Integral Field Unit for Mapping Feedback in Galaxies.* arXiv:2607.23020.
4. Rinaldi, P. et al., including **Ma, Z.** (2026), *The Way We Tally Becomes the Tale: The Impact of Selection Strategies on the Inferred Evolution of Little Red Dots Across Cosmic Time.* arXiv:2604.07138.
5. Zhu, Y. et al., including **Ma, Z.** (2026), *Clumps in High-Redshift Galaxies: Mass Scaling and Radial Trends from JADES.* ApJ, 1000. DOI.
6. Zhu, Y. et al., including **Ma, Z.** (2025), *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$.* ApJL, 995, L5. arXiv:2509.00153.
7. Zhang, J. et al., including **Ma, Z.** (2025), *Abundant Population of Broad Hα Emitters in the GOODS-N Field Revealed by CONGRESS, FRESCO, and JADES.* ApJ. arXiv:2505.02895.
8. Sun, F. et al., including **Ma, Z.** (2025), *Slitless Areal Pure-Parallel High-Redshift Emission Survey (SAPPHIRES): Early Data Release of Deep JWST/NIRCam Images and Spectra in the MACS J0416 Parallel Field.* arXiv:2503.15587.
9. Fujimoto, S. et al., including **Ma, Z.** (2025), *Primordial Rotating Disk Composed of at Least 15 Dense Star-Forming Clumps at Cosmic Dawn.* Nature Astronomy, 9, 1553–1567. DOI.

Presentations

- **NOIRLab FLASH Talk**, Tucson, AZ, USA, November 13, 2026. *Upcoming Talk.*
- **JWST 2026 Conference**, Tokyo, Japan, 2026. *Talk.*
- **JADES Collaboration Meeting**, Santa Cruz, CA, USA. *Talk.*
- **247th Meeting of the American Astronomical Society**, Phoenix, AZ, USA, January 2026. *Poster Presentation.*

References

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